SAPIENZA
UNIVERSITÀ DI ROMA

MSc Cybersecurity

Course in Economics of Technology and Management



Sustainable farming with EYPE Greenhouse

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Introduction

The modern era presents us with unprecedented challenges regarding the conscious use of resources, directly aligning with **SDG 12: Responsible Consumption and Production**. Currently, traditional agriculture accounts for roughly **70% of global freshwater** withdrawals. In this context, it is crucial to pay particular attention to the conscious use of soil and water. The latter, being an increasingly scarce commodity, requires the most prudent and informed consumption. Adopting innovative solutions like hydroponic greenhouses addresses this exact issue, allowing for up to a **90% reduction in water usage** and proving that we can produce more with less.



Introduction

Modern technology offers sophisticated tools to reduce waste without neglecting economic costs. In this analysis, I present **EYPE** (*Enjoy Your Plant Everywhere*), a **hydroponic greenhouse** that allows you to grow vegetables and greens while saving water and achieving economic benefits compared to traditional farming techniques. Beyond resource efficiency, EYPE maximizes spatial capacity. By optimizing the growing environment, hydroponic systems can achieve crop yields up to **10 times higher** per square meter than conventional soil agriculture, making it a highly scalable solution for localized food production.



Research objective

The aim of the research is to demonstrate how the use of the EYPE hydroponic greenhouse can lead to significant economic savings, while ensuring water savings and low operating costs.

Methodology

The methodology I used to conduct this economic analysis is the **Discounted Cash Flow Analysis**.

ECONOMIC INDICATORS

- **Key economic metrics:** Project assessed based on several economic indicators (NPV, PBT, DPBT, IRR, PI).

INPUT DATA

- **CAPEX:** Structural components, sensors, actuators, assembly and setup.
- **OPEX:** Energy, chemical products, maintenance and software hosting.
- **Benefits:** Produce procurement savings and resource efficiency (~90% water reduction).

ANALYSIS & SIMULATION

- **Sensitivity and Scenario analysis:** Testing optimistic and pessimistic variation in value of critical variables.
- **Break-even analysis:** Break-even point under variation of greens production or their pricing, and structural costs.
- **Risk analysis:** Run two Monte Carlo simulations evaluating the robustness of the project under volatile market conditions.

Economic indicators

1 NET PRESENT VALUE

The total net profit of the project, expressed in today's value.

$$\sum_{t=0}^{N} \frac{F_t}{(1+r)^t}$$

2 PAYBACK TIME

The exact time required to fully recover the initial investment.

$$\sum_{t=0}^{PBT} F_t = 0$$

3 DISCOUNTED PAYBACK TIME

The time needed to break even, factoring in the time value of money.

$$\sum_{t=0}^{DPBT} \frac{F_t}{(1+r)^t} = 0$$

4 INTERNAL RATE OF RETURN

The annualized percentage return generated by the investment.

$$\sum_{t=0}^{N} \frac{F_t}{(1+IRR)^t} = 0$$

5 PROFITABILITY INDEX

The amount of value created for every single Euro invested.

$$PI = \frac{NPV + Investment}{Investment}$$

Input data

For the input data, I considered a total **initial investment (CAPEX) of 950€**, which is the production cost of the greenhouse. The costs are divided as follow:

- Structural components (aluminum profile, Lexan, abet wood, 3D printing).
- Actuators (pumps, solenoid valves, RGB LEDs).
- Assembly and software setup (amortized per unit).

For the **operating costs (OPEX)** i considered a flat annual value of **420€**, divided among:

- Electricity (consumption of 876 kWh/year at 0.25 €/kWh).
- Chemical products.
- Maintenance and hosting of an ASP.NET server on cloud.

Data taken starting from the official EYPE documentation

Input data

For the benefits I calculated **annual savings** for a total of **1250€** considering:

- Produce procurement of vegetables that cannot be grown in traditional farming in certain locations or period of the year.
- Water savings (90% reduction compared to traditional agricolture: 22.5 m^3 at 1.50 €/m^3).
- Soil and fertilizer avoided costs for traditional farming.

To carry out the analysis, I considered a **time period of 5 years** and a **discount rate** (opportunity cost of capital) **of 5%,** which is a typical value for similar projects in the green economy area, and it also reflects the average return of low-risk investments.

Data taken starting from the official EYPE documentation

Input data

INITIAL INVESTMENT	
DESCRIPTION	**VALUE**
Structural components	350 €
Sensors, controllers, mini PC	300 €
Actuators	150 €
Assembly & Software setup	150 €
Total	950 €

ANNUAL OPERATING COSTS	
DESCRIPTION	**VALUE**
Electricity (876 kWh/year at €0.25/kWh)	219 €/year
Chemical products	100 €/year
Maintenance & Cloud Server Hosting	101 €/year
Total	420 €/year

Input data

ANNUAL BENEFITS	
DESCRIPTION	**VALUE**
Produce procurement savings (100 kg herbs at €12/kg)	1200 €/year
Water savings (90% reduction vs traditional: 22.5 m^3 @ €1.50/m^3)	34 €/year
Soil and fertilizer avoided costs	16 €/year
Total	1250 €/year

Economic analysis: results

Baseline scenario

The analysis highlights the exceptional profitability of the EYPE project and a low risk profile: the initial investment is **fully recovered in** just over **1 year and 1 month** (**1 year and 2 months** considering **discounting**), generating a significantly positive **NPV of € 2643** over a five-year horizon. With an I**RR of 83%** and a **Profitability Index of 3.8**, the model demonstrates extraordinary efficiency in creating net value.

KEY INDICATOR	VALUE
NPV	2643 €
PBT	1 year 1 month 22 days
DPBT	1 year 2 months 18 days
IRR	83%
PI	3,8

Sensitivity analysis

To perform an effective sensitivity analysis, I varied the project's critical cost items, which influence its profitability and payback time. The first variable is the one relating to the **structural costs** of EYPE.

- With a **pessimistic 20% increase** in costs, a **very high NPV** is still achieved, and the **DPBT** is set over a time horizon **of less than 1 year and 4 months**. Therefore, over the five-year time horizon considered, the project has a very rapid payback and guarantees net cash flows for almost its entire useful life.
- For completeness, the **optimistic case with a 20% decrease** in structural costs is also evaluated, further strengthening the investment's profitability.

INCREASE	NPV	DPBT	IRR	PI
0% (Baseline)	2643 €	1,2 years	83%	3,8
+20%	2573 €	1,3 years	77%	3,5
-20%	2713 €	1,1 years	91%	4,1

Sensitivity analysis

To perform an effective sensitivity analysis, I varied the project's critical cost items, which influence its profitability and payback time. The second variable is that relating to **electricity costs**.

- With a **pessimistic cost increase of 20%**, an **NPV of €2454** is obtained and a **DPBT of less than 1 year and 4 months**. Therefore, over the five-year timeframe considered, the project has a very rapid payback and guarantees net cash flows for almost its entire useful life.
- For completeness, the **optimistic case with a 20% decrease** in structural costs is also evaluated, further strengthening the investment's profitability.

INCREASE	NPV	DPBT	IRR	PI
0% (Baseline)	2643 €	1,2 years	83%	3,8
+20%	2454 €	1,3 years	78%	3,6
-20%	2833 €	1,1 years	88%	4,0

Sensitivity analysis

To perform an effective sensitivity analysis, I varied the project's critical cost items, which influence its profitability and payback time. The third variable is that relating to the **savings generated by product procurement**.

- With an **optimistic 20% increase** in vegetable costs, we obtain an **NPV of €3683**: significantly higher because the savings this time are greater. Furthermore, in this case, **the payback period is less than 11 months**.
- In the **pessimistic case of a 20% decrease** in costs, the savings would be lower, therefore, we obtain a **lower NPV** and a **DPBT of approximately 1 year and 8 months**.

INCREASE	NPV	DPBT	IRR	PI
0% (Baseline)	2643 €	1,2 years	83%	3,8
+20%	3683 €	0,9 years	110%	4,9
-20%	1604 €	1,7 years	55%	2,7

Sensitivity analysis

To perform an effective sensitivity analysis, I varied the project's critical cost items, which influence its profitability and payback time. The fourth analysis is performed by varying the **discount rate**.

- A discount rate of **20%** (**high risk**) and **1%** (**very low risk**) are considered. In both cases, the **NPV is very high**, and the project always **recovers its investment in less than 1 year and 5 months** (worst-case), confirming the project's high profitability even in high-risk economic environments.

INCREASE	NPV	DPBT	IRR	PI
5% (Baseline)	2643 €	1,2 years	83%	3,8
20%	1532 €	1,4 years	83%	2,6
1%	3078 €	1,2 years	83%	4,2

Scenario analysis

The scenario analysis was conducted by considering the variables that most significantly impact the project's profitability. Specifically, the **structural costs** of the greenhouse, **electricity**, and the market **price of vegetables** were simultaneously varied, defining a **pessimistic** and an **optimistic** scenario. This allows us to assess the robustness of the economic model, projecting its performance under different market conditions.

	STRUCTURAL	ENERGY	HERBS
BASELINE	+0% (Baseline)	+0% (Baseline)	12 €/kg (Baseline)
PESSIMISTIC	+15%	+30%	10 €/kg
OPTIMISTIC	-10%	-10%	14 €/kg

Scenario analysis

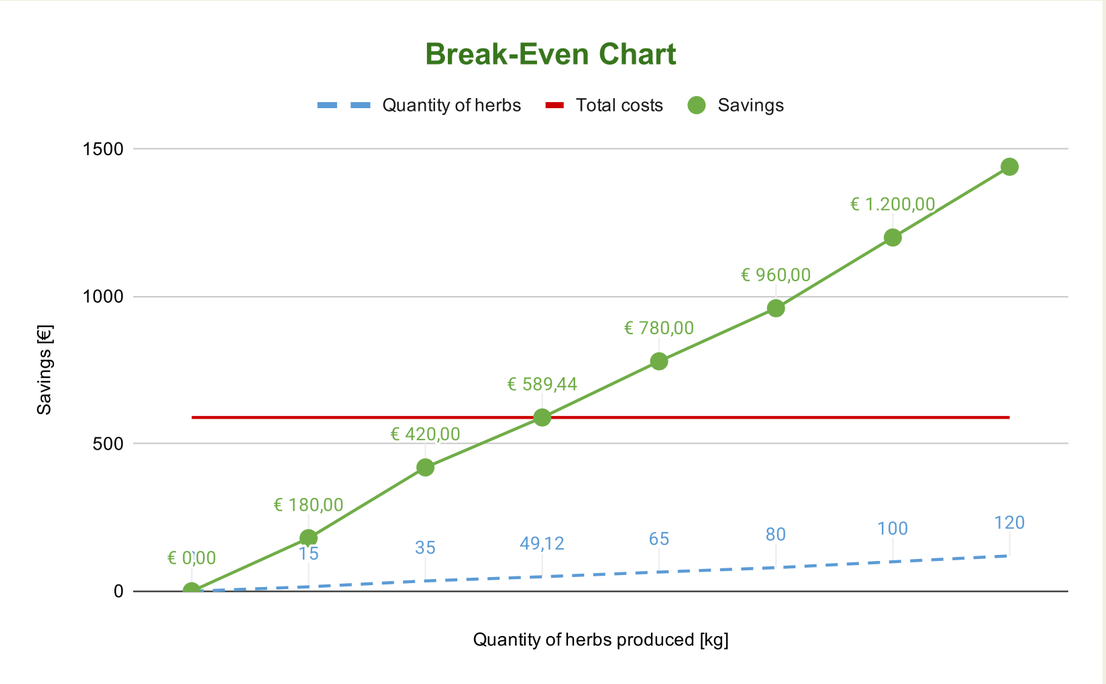
The **NPV values are very high**, further reinforcing the positive investment outlook for the EYPE project. Furthermore, we can estimate a payback period of nearly **1 year and 11 months in the worst-case scenario**, meaning the project would generate profits for most of its life even in an adverse economic context, confirming the project's profitability.

	NPV	DPBT	IRR	PI
BASELINE	2643 €	1,2 years	83%	3,8
PESSIMISTIC	1441 €	1,9 years	48%	2,4
OPTIMISTIC	3639 €	0,9 years	112%	5,0

Break-even analysis

The break-even analysis was carried out with respect to the savings achieved by vegetables and to the structural costs, which are the two variables that most influence the project's profitability.

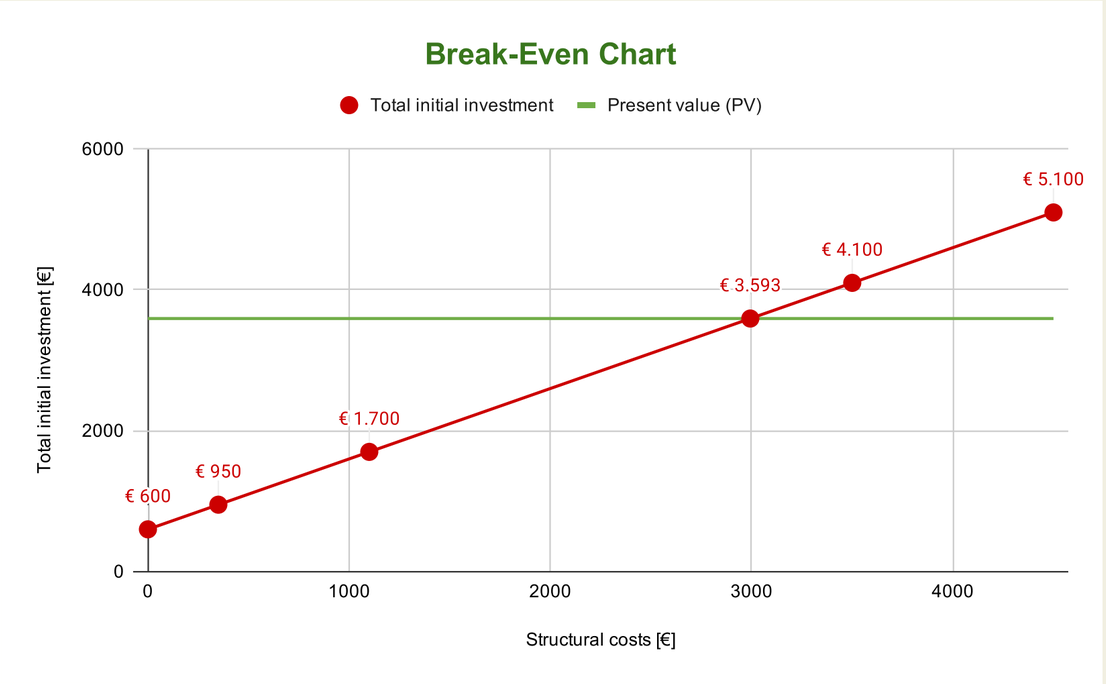
Specifically, the **break-even point** (**BEP**), over the 5-year time horizon considered, is reached with a **minimum of 49.1 kg of herbs and greens** produced at a cost of €12/kg, or with 100 kg produced at a **minimum cost of €5.89**. The resulting numbers indicate the project's extreme robustness, as these quantities are easily achievable in a hydroponic context.



Break-even analysis

The break-even analysis was carried out with respect to the savings achieved by vegetables and to the structural costs, which are the two variables that most influence the project's profitability.

Regarding the structural costs, **to nullify the profit** generated by the greenhouse over 5 years, **a cost of €2993.47 would be necessary**, corresponding to an **increase of ~855%.** This is an unrealistic scenario and therefore confirms the project's resilience even to the volatility of the cost of raw materials and semi-finished products.



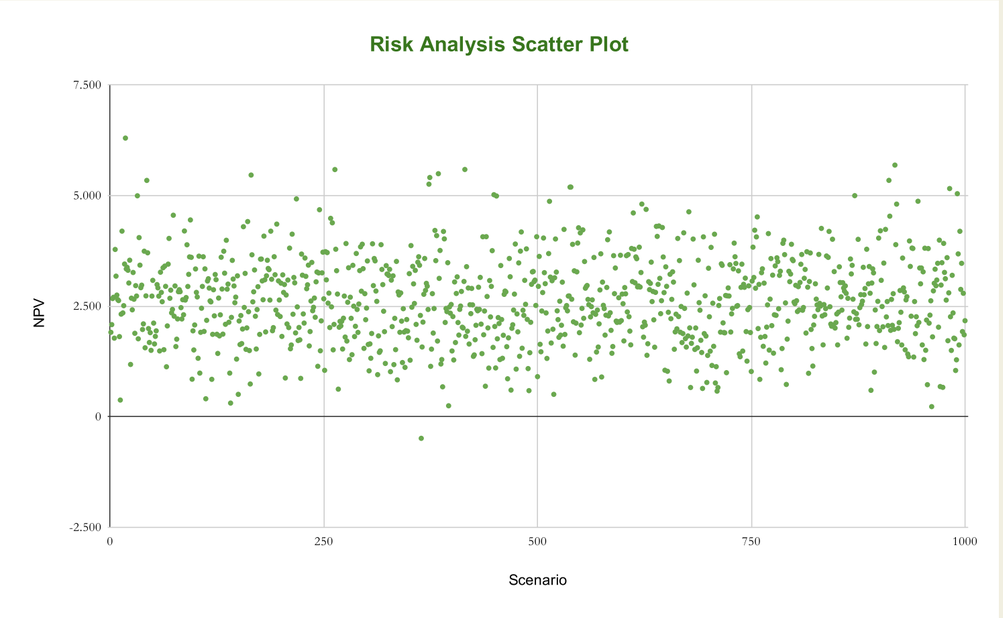
Risk analysis

The sensitivity analyses and scenario analyses both yielded very positive results for the EYPE greenhouse. To conduct a comprehensive evaluation of the project, two risk analyses were conducted, taking into account the savings resulting from vegetable cultivation and the change in the greenhouse's structural costs.

For the first analysis, a **Monte Carlo simulation** was performed with the following input data:

- Initial investment: average value of €950 with a 10% standard deviation.
- Annual operating costs: average value of €420 with a 10% standard deviation.
- Quantity of herbs produced: average value of €100 with a 10% standard deviation.
- Price of herbs: average value of 12 €/kg with a 15% standard deviation.

By simulating 1000 different scenarios, an **average success probability of 99.8%** is obtained, which indicates the project's extreme resilience to market fluctuations regarding the cost of vegetables or their production.

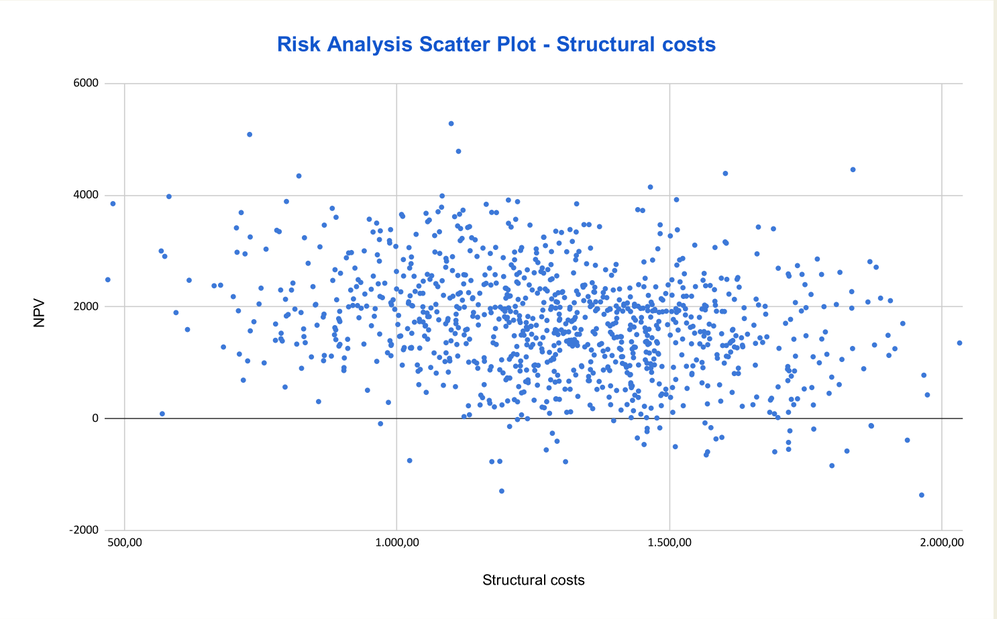


Risk analysis

For the second analysis, a **Monte Carlo simulation** was performed considering a much higher average value of structural costs.

- Structural costs: average value of €1300 with a 20% standard deviation.
- Other initial costs: average value of €600 with a 20% standard deviation.
- Annual operating costs: average value of €420 with a 20% standard deviation.
- Annual savings: average value of €1250 with a 15% standard deviation.

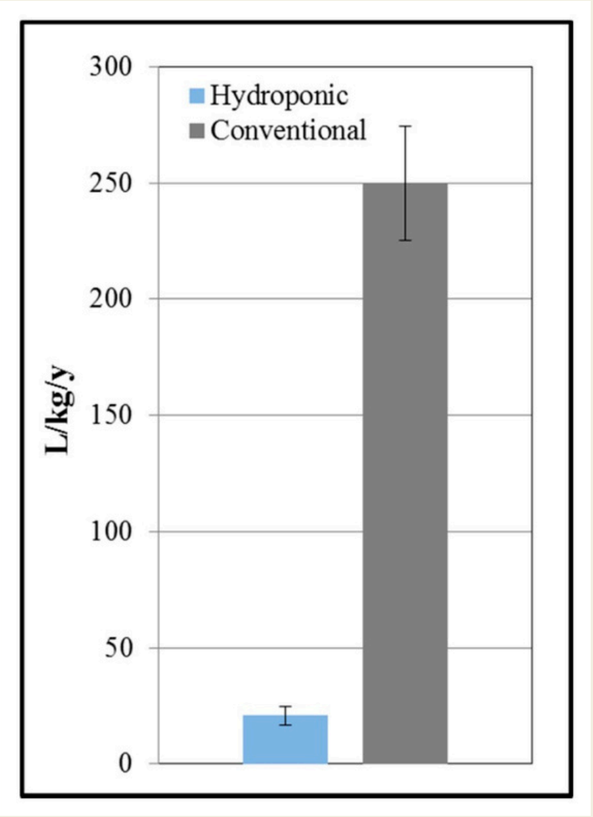
Simulating 1000 different scenarios, an **average probability of success of 95.9%** is obtained. Therefore, even significantly increasing the average value of structural costs to account for any possible initial underestimation, the project proves robust and potentially profitable with an extremely low risk profile.



Hydroponic greenhouses: benefits and drawbacks

Considerations on sustainability

We have seen that EYPE is very economically sound, but is the choice to use a hydroponic greenhouse truly always sustainable? In the scientific community, the real debate about hydroponic greenhouses revolves around a major environmental trade-off: they save a lot of water and soil, but **consume much more electricity than traditional agriculture**. Scientific literature warns that hydroponic greenhouses risk creating a **CO_2 emissions issue** due to the electricity consumed for pumps and sensors. Therefore, the real future challenge for EYPE, to be in line not only with SDG 12 (Responsible Consumption and Production) but also with **SDG 13 (Climate Action)**, will be to cover electricity costs with **renewable energy sources**.

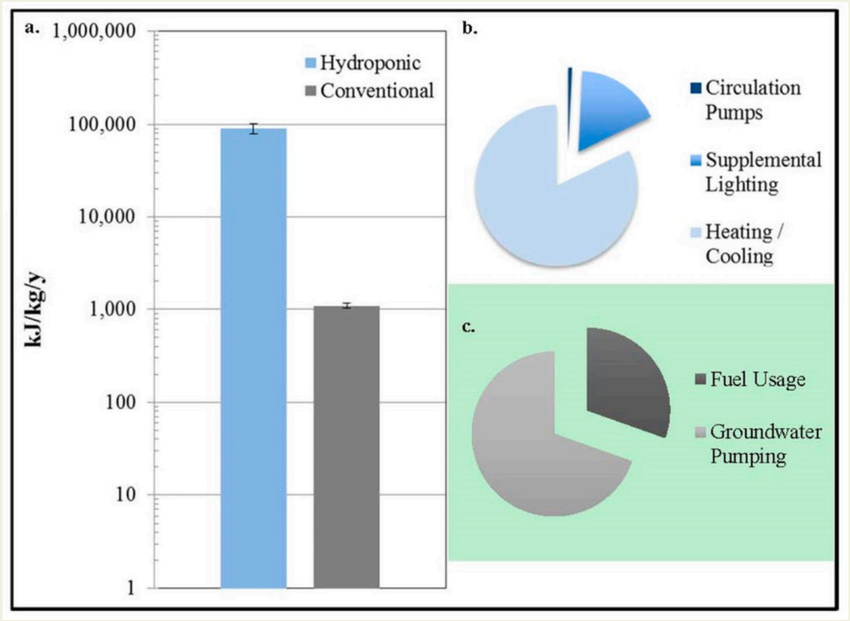


Water consumption: Hydroponic vs. Conventional agricolture.

Considerations on sustainability

The study by Barbosa *et al.* (2015) *"Comparison of Land, Water, and Energy Requirements of Lettuce Grown Using Hydroponic vs. Conventional Agricultural Methods"* confirms that **hydroponics reduces water consumption by over 90%** while using very little space, but highlights a technical criticality: the **energy required** for water recirculation and climate control is up to **80 times higher** than traditional agriculture, risking greater pollution if powered by fossil fuels. However, to adequately assess the system's true sustainability, a broader life cycle perspective must be adopted.

The study by Sanjuan-Delmás *et al.* (2018) *"Environmental assessment of an integrated rooftop greenhouse for food production in cities"* demonstrates that moving food production to urban contexts **eliminates waste resulting from logistics**. This shortening of the agri-food supply chain eliminates so-called "food miles", **largely compensating the environmental impact** resulting from electricity consumption. The hydroponic greenhouse model is positioned exactly within this "zero-kilometer" sustainability perspective.



Energy consumption: Hydroponic vs. Conventional agricolture.



Conclusions

Ultimately, EYPE's economic analysis demonstrates that technological innovation overcomes the traditional tradeoff between ecology and profit. With a 90% reduction in water use and a yield per square meter up to 10 times higher than conventional agriculture, the project positions itself as a **highly scalable solution**. Economically, the model guarantees an **extraordinary safety margin**: the ability to maintain net profitability even in front of a potential 855% increase in material costs and the 95.9% success rate in stress tests confirm that hydroponic efficiency today represents a significant **competitive advantage**.

Key findings

ECONOMIC VIABILITY

The EYPE project demonstrates excellent profitability, recovering the initial investment extremely quickly (**DPBT of ~1 year and 2 months**) and generating a **high NPV** (**€2643**) over the 5-year horizon.

RISK RESILIENCE

Monte Carlo simulations and scenario and sensitivity analyses confirm the model's robustness, ensuring a **success rate above 95%** even when simulating highly volatile market conditions or increases in structural and energy costs.

SUSTAINABILITY

EYPE fully **meets SDG 12** by optimizing resource efficiency and promoting responsible consumption: the high energy requirements typical of hydroponic systems are compensated by the **90% reduction in water consumption** and the zero-food miles transportation logistics guaranteed by the **zero-kilometer model**. Future developments will focus on **integrating renewable energy sources** to cover operating energy costs, thus aligning the project with **SDG 13**.

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Thank you

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